VAE & Latent Variable Models Quiz: Ready-to-Use HTML (38 Questions, 14 Groups)

Below is the complete HTML file for your quiz engine: 38 questions in 14 groups, running from easy recall to mostly hard transfer and deduction questions, with 3 tries per group and an answer bubble beside each question.

TL;DR

- The file has 38 questions in 14 topic groups: 3 easy, 12 medium and 23 hard (transfer or deduction). They cover slides 41–78, including the figures (GMM ellipses, the EM-bound plot, the KL-ellipse packing, the MNIST latent scatter plots, the VQ layer and the VQ-VAE vs dVAE table) and the speaker notes.
- Every numeric answer was checked by hand. Examples: closed-form KL of 0.5 / 0.568 / 1.068; ln 512 ≈ 6.238; reparameterised gradient 0.06; VQ distance matrix 8192×512; 9216 bits and a 393,216/9,216 = 42.67× compression (van den Oord et al. §4.2 report "≈ 42.6 in bits"). These match Kingma & Welling (Appendix B), van den Oord et al. (VQ-VAE) and Jang et al. (Gumbel-Softmax).
- The file contains only HTML plus an inline config script. It loads the existing `youtube-quiz-engine.css` / `.js` and follows the layout rules: first question at y≈9–10, labels directly on top of their inputs, and the answer bubble on the right at the same height as its question.

The HTML file (save as `VAE_latent_variable_models_quiz.html`)

```html
```

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8" />
<meta name="viewport" content="width=device-width, initial-scale=1.0" />
<title>Latent Variable Models &amp;amp; VAEs — Quiz</title>
<link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/katex@0.16.21/dist/katex.min.cs
<script defer src="https://cdn.jsdelivr.net/npm/katex@0.16.21/dist/katex.min.js"></script
<link rel="stylesheet" href="youtube-quiz-engine.css" />
</head>
<body>
<div id="header">Machine Learning — Latent Variable Models &amp;amp; VAEs Quiz</div>
<div id="quiz-frame">
  <div id="video-wrap">
    <iframe id="yt-player" allow="autoplay"></iframe>
    <div id="overlay-layer"></div>
    <div id="pause-notice">PAUSED &mdash; answer the question</div>
  </div>
</div>
<div id="footer">Click ▶ to start · Tab between fields · Enter to submit</div>

<script>
var VIDEO_ID = '';
var BACKGROUND = { enabled: true, color: '#ffffff', autoPlay: false };
var DEFAULT_TIME_GAP = 4;
var POST_QUIZ_DURATION = 12;
var MOODLE = { enabled: false };

var BUBBLE_DEFAULTS = { style:'bubble', tail:'left', bg:'rgba(255,248,200,0.97)', borderC
function mkBubble(text, y, opts){ var b={x:70,y:y,w:28,text:text}; for(var k in BUBBLE_DE

/* small inline helpers for cloze blanks (part of this HTML, not a new JS file) */
function tf(a){ return {blank:true, type:'select', options:['True','False'], ans:a, w:70}
function num(a,tol,w){ return {blank:true, ans:String(a), tol:tol, w:(w||60)}; }
function sel(opts,a,w){ return {blank:true, type:'select', options:opts, ans:a, w:(w||130

var GROUPS = [

/* ==================== G1 ==================== */
{ label:'GMM & Latent Variable Basics', maxTries:3,
  questions:[
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    { type:'radio', x:5, y:10, w:62, fs:11,
      question:'<b>Q1.</b> In the GMM Bayes net $z\\to x$ with $z\\sim\\mathrm{Cat}(\\pi_
      choices:['$p(x\\mid z=k)$ — the component likelihood',
               '$p(z=k\\mid x)\\propto \\pi_k\\,\\mathcal N(x;\\mu_k,\\Sigma_k)$ — the po
               '$p(z=k)=\\pi_k$ — the mixing weight',
               '$p(x)=\\sum_k \\pi_k\\,\\mathcal N(x;\\mu_k,\\Sigma_k)$ — the marginal'],
      ans:1 },
    { type:'cloze', x:5, y:40, w:62, fs:11, lineSpacing:1.7,
      parts:['<b>Q2.</b> True or False?<br>(a) Maximum-likelihood fitting of a GMM is conv
             '<br>(b) The learned $z$ of an unsupervised GMM/VAE is guaranteed to be humar
             '<br>(c) A single Gaussian cannot fit the two-cluster data of slide 42, but a
    { type:'cloze', x:5, y:64, w:62, fs:11, lineSpacing:1.6,
      parts:['<b>Q3.</b> A GMM with $K=3$ full-covariance components in $\\mathbb R^2$ ha
             ' free parameters (means + covariances + mixing weights).'] }
  ],
  bubbles:[
    mkBubble('<b>Q1 — B.</b> Clustering = inferring the hidden cause $p(z{=}k|x)$ by Baye
    mkBubble('<b>Q2 — F, F, T.</b> The GMM likelihood is highly non-convex (local minima,
    mkBubble('<b>Q3 — 17.</b> Means 3×2=6; each symmetric 2×2 covariance has 3 → 9; weight
  ]
},

/* ==================== G2 ==================== */
{ label:'VAE as an Infinite Mixture', maxTries:3,
  questions:[
    { type:'radio', x:5, y:10, w:62, fs:11,
      question:'<b>Q4.</b> Slide 43: $z\\sim\\mathcal N(0,1)$, $p(x|z)=\\mathcal N(\\mu_\
      choices:['It is Gaussian, because a Gaussian prior and Gaussian likelihood are conju
               'It is a continuous (infinite) mixture of Gaussians — generally non-Gaussia
               'It is a mixture of exactly $K$ Gaussians, where $K$ is the decoder width'
               'It equals $p(x\\mid z=0)$ because $z=0$ is the prior mode'],
      ans:1 },
    { type:'cloze', x:5, y:42, w:62, fs:11, lineSpacing:1.6,
      parts:['<b>Q5.</b> With $\\Sigma_\\theta(z)=\\mathrm{diag}(\\exp(\\sigma(Bz+d)))$, 
             num(1,0.01,50), ' , ', num(2.718,0.02,60), ' ).'] },
    { type:'checkbox', x:5, y:56, w:62, fs:11,
      question:'<b>Q6.</b> Select ALL correct statements about deep latent variable models
      choices:['Training is unsupervised: only $x$ is observed',
               'The prior $p(z)$ must stay fixed; it can never be learned',
               'Sampling is ancestral: $z\\sim p(z)$, then $x\\sim p(x|z)$',
               'The NN replaces the GMM lookup table $k\\mapsto(\\mu_k,\\Sigma_k)$ by a f
      ans:[0,2,3] }
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  ],
  bubbles:[
    mkBubble('<b>Q4 — B.</b> $p(x)=\\int p(z)\\,\\mathcal N(x;\\mu_\\theta(z),\\Sigma_\\t
    mkBubble('<b>Q5 — (1, e≈2.718).</b> Sigmoid ∈ (0,1) ⇒ exp ∈ ($e^0$, $e^1$). Deductio
    mkBubble('<b>Q6 — A, C, D.</b> B is false: priors can be learned (hierarchical VAEs, 
  ]
},

/* ==================== G3 ==================== */
{ label:'Evaluating p(x): Monte Carlo & Importance Sampling', maxTries:3,
  questions:[
    { type:'cloze', x:5, y:10, w:62, fs:11, lineSpacing:1.6,
      parts:['<b>Q7.</b> With 20 binary latents, exact $p(x)=\\sum_z p(x,z)$ needs ', num
             ' terms; with 30 binary latents ≈ ', num(1.07,0.02,55), ' billion terms.'] }
    { type:'radio', x:5, y:26, w:62, fs:11,
      question:'<b>Q8.</b> Naive MC $p(x)\\approx|\\mathcal Z|\\tfrac1K\\sum_k p(x,z^{(k)
      choices:['the estimator is biased for any finite $K$',
               'for almost all random $z$, $p(x|z)\\approx0$; the few $z$ that explain $x
               'uniform distributions cannot be sampled on a computer',
               'the logarithm of the estimator is always $+\\infty$'],
      ans:1 },
    { type:'radio', x:5, y:57, w:62, fs:11,
      question:'<b>Q9.</b> Importance sampling: $p(x)=\\mathbb E_{z\\sim q}[p(x,z)/q(z)]$
      choices:['$q(z)=p(z)$, the prior','$q(z)$ uniform','$q(z)=p(z\\mid x)$, the true po
      ans:2 }
  ],
  bubbles:[
    mkBubble('<b>Q7 — 1,048,576; ≈1.07.</b> $2^{20}$ and $2^{30}$ configurations: exponen
    mkBubble('<b>Q8 — B.</b> Unbiased but huge variance: $p(z|x)$ concentrates in a tiny 
    mkBubble('<b>Q9 — C.</b> With $q=p(z|x)$, $p(x,z)/q(z)=p(x)$ for every $z$: constant 
  ]
},

/* ==================== G4 ==================== */
{ label:'ELBO, Jensen & the KL Gap', maxTries:3,
  questions:[
    { type:'radio', x:5, y:10, w:62, fs:11,
      question:'<b>Q10.</b> Why does $\\log\\mathbb E_q\\!\\left[\\tfrac{p(x,z)}{q(z)}\\r
      choices:['Because log is convex (Jensen)','Because log is concave (Jensen)','Becaus
      ans:1 },
    { type:'cloze', x:5, y:38, w:62, fs:11, lineSpacing:1.7,
      parts:['<b>Q11.</b> For one image $\\log p_\\theta(x)=-100$ nats and ELBO $=-104.5$
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            '.<br>A better encoder (θ unchanged) reduces this gap to 1.2; the new ELBO i
    { type:'checkbox', x:5, y:56, w:62, fs:11,
      question:'<b>Q12.</b> The encoder ignores $x$ and outputs $q(z|x)=p(z)=\\mathcal N(
      choices:['$D_{KL}(q\\|p(z))=0$','The ELBO equals $\\mathbb E_{p(z)}[\\log p_\\theta
      ans:[0,1,2] }
  ],
  bubbles:[
    mkBubble('<b>Q10 — B.</b> For concave $f$: $f(\\mathbb E[w])\\ge\\mathbb E[f(w)]$. Eq
    mkBubble('<b>Q11 — 4.5; −101.2.</b> $\\log p(x)=\\text{ELBO}+D_{KL}(q\\|p(z|x))$. Cha
    mkBubble('<b>Q12 — A, B, C.</b> Any $q$ gives a valid bound. The gap $D_{KL}(p(z)\\|p
  ]
},

/* ==================== G5 ==================== */
{ label:'EM-like Bounds & Stochastic VI', maxTries:3,
  questions:[
    { type:'radio', x:5, y:10, w:62, fs:11,
      question:'<b>Q13.</b> Slide 48: the red bound $\\mathcal L(x;\\theta,\\phi_1)$ touc
      choices:['Touching means $q$ = exact posterior at $\\theta_n$, so $\\log p(\\theta_
               'The bound is convex in $\\theta$',
               'The orange bound always lies above the red one',
               'Nothing — maximising a bound can never improve the likelihood'],
      ans:0 },
    { type:'ordering', x:5, y:45, w:62, fs:11,
      question:'<b>Q14.</b> Order the steps of (non-amortised) stochastic variational inf
      items:['Update $\\theta\\leftarrow\\theta+\\eta\\nabla_\\theta\\mathcal L(x^i;\\the
             'Initialise $\\theta$ and $\\phi^1,\\dots,\\phi^M$',
             'Repeat $\\phi^i\\leftarrow\\phi^i+\\eta\\nabla_{\\phi^i}\\mathcal L$ until
             'Sample a data point $x^i$ from $\\mathcal D$',
             'Compute $\\nabla_\\theta\\mathcal L(x^i;\\theta,\\phi^{i*})$ ($\\phi$ fixed
      ans:[1,3,2,4,0] }
  ],
  bubbles:[
    mkBubble('<b>Q13 — A.</b> The EM/MM argument. With an approximate, amortised $q$ the
    mkBubble('<b>Q14 — init → sample $x^i$ → optimise $\\phi^i$ → $\\nabla_\\theta$ → upd
  ]
},

/* ==================== G6 ==================== */
{ label:'Reparameterization Trick', maxTries:3,
  questions:[
    { type:'cloze', x:5, y:10, w:62, fs:11, lineSpacing:1.8,
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      parts:['<b>Q15.</b> $z=\\mu+\\sigma\\varepsilon$ with $\\mu=0.5,\\ \\sigma=2,\\ \\v
             ', $\\partial z/\\partial\\mu=$ ', num(1,0.01,45), ', $\\partial z/\\partial
             '.<br>For $r(z)=z^2$: $\\partial r/\\partial\\sigma=$ ', num(0.06,0.005,55),
    { type:'radio', x:5, y:28, w:62, fs:11,
      question:'<b>Q16.</b> For which $q(z;\\phi)$ does plain reparameterisation $z=g(\\v
      choices:['Gaussian $\\mathcal N(\\mu,\\sigma^2)$','Laplace (location–scale family)'
      ans:3 },
    { type:'cloze', x:5, y:58, w:62, fs:11, lineSpacing:1.5,
      parts:['<b>Q17.</b> Slide 49 flags $\\nabla_\\phi\\mathbb E_{q(z;\\phi)}[\\log p(z,
             {blank:true, type:'open', w:520, h:55,
              ans:'The sampling distribution itself depends on phi, so we cannot just samp
              placeholder:'Write your answer here. It will be evaluated by AI…'}] }
  ],
  bubbles:[
    mkBubble('<b>Q15 — −0.1; 1; −0.3; 0.06.</b> $\\partial z/\\partial\\sigma=\\varepsilo
    mkBubble('<b>Q16 — D.</b> A discrete sample is a step function of ϕ (zero gradient al
    mkBubble('<b>Q17 — model answer.</b> The <i>measure</i> depends on ϕ. With $z=\\mu+\\
  ]
},

/* ==================== G7 ==================== */
{ label:'Amortized Inference & the Amortization Gap', maxTries:3,
  questions:[
    { type:'radio', x:5, y:10, w:62, fs:11,
      question:'<b>Q18.</b> Amortisation replaces per-point $\\phi^i$ with an encoder $f_
      choices:['always higher, because the encoder shares statistical strength',
               'at most equal; the difference is the <i>amortization gap</i>',
               'exactly equal for any encoder architecture',
               'undefined, because an amortised $q$ is not a distribution'],
      ans:1 },
    { type:'label', x:5, y:44, w:62, fs:11,
      text:'<b>Q19.</b> Match each quantity / remedy with the gap it concerns ($q^*$ = be
    { type:'match', x:5, y:49, w:520, fs:10, dotSize:9, dotGap:7, colGap:40,
      left:['$\\log p(x)-\\mathcal L[q^*]$','$\\mathcal L[q^*]-\\mathcal L[q_\\phi]$','Ri
      right:['Amortization gap','Approximation gap'],
      ans:[1,0,1,0] }
  ],
  bubbles:[
    mkBubble('<b>Q18 — B.</b> $f_\\lambda$ can only reach a subset of per-point optima. C
    mkBubble('<b>Q19 — Approx, Amort, Approx, Amort.</b> The family limits how close <i>a
  ]
},
```

```
/* ==================== G8 ==================== */
{ label:'ELBO Decomposition & Closed-Form KL', maxTries:3,
  questions:[
    { type:'cloze', x:5, y:10, w:62, fs:11, lineSpacing:1.8,
      parts:['<b>Q20.</b> Using $D_{KL}=\\tfrac12\\sum_j(\\mu_j^2+\\sigma_j^2-\\log\\sigm
             '; $\\mu=0,\\sigma=e^{-1}$: ', num(0.568,0.005,60),
             '<br>2-D code $\\mu=(1,0)$, $\\sigma=(1,e^{-1})$: ', num(1.068,0.01,60), ' n
    { type:'slider', x:5, y:30, w:62, fs:11,
      question:'<b>Q21.</b> Loss $=\\|x-\\mu_\\theta(z)\\|^2+D_{KL}$ equals the negative
      min:0, max:2, step:0.05, ans:0.5, tol:0.05, unit:'' },
    { type:'cloze', x:5, y:52, w:62, fs:11, lineSpacing:1.7,
      parts:['<b>Q22.</b> True or False?<br>(a) With a fixed $\\mathcal N(0,I)$ prior, th
             '<br>(b) $\\mathbb E_q[\\log p(x,z)]+H(q)$ and $\\mathbb E_q[\\log p(x|z)]-D
             '<br>(c) The reconstruction term alone pushes the encoder σ toward 1. ', tf(
  ],
  bubbles:[
    mkBubble('<b>Q20 — 0.5; 0.568; 1.068.</b> ½(1+1−0−1)=0.5; ½(0+e⁻²+2−1)=0.568. Diagonal
    mkBubble('<b>Q21 — 0.5.</b> $-\\log p(x|z)=\\|x-\\mu\\|^2/(2s^2)+$const, equal to MSE
    mkBubble('<b>Q22 — T, T, F.</b> θ appears only in $p(x|z)$. $\\log p(x,z)=\\log p(z)+
  ]
},

/* ==================== G9 ==================== */
{ label:'Reading the Latent-Space Figures', maxTries:3,
  questions:[
    { type:'label', x:5, y:9, w:62, fs:11,
      text:'<b>Q23.</b> Match each training setting (2-D MNIST latents, slide 57) to its
    { type:'match', x:5, y:14, w:520, fs:10, dotSize:9, dotGap:7, colGap:40,
      left:['Reconstruction loss only','KL loss only','Reconstruction + KL (standard VAE)
      right:['Separated, irregular clusters with large gaps; prior samples often decode t
             'All classes overlap in one blob at the origin; decoder outputs an "average"
             'Clusters packed near the origin with smooth transitions between classes'],
      ans:[0,1,2,1] },
    { type:'radio', x:5, y:50, w:62, fs:11,
      question:'<b>Q24.</b> In the KL figure (red ellipses inside the blue $\\mathcal N(0
      choices:['KL weight too large','KL weight ≈ 0 — the encoder behaves like a determin
      ans:1 }
  ],
  bubbles:[
    mkBubble('<b>Q23 — A, B, C, B.</b> β→∞ makes KL dominate, so $q(z|x)\\to p(z)$ for ev
    mkBubble('<b>Q24 — B.</b> Without KL pressure σ→0 (dots) and means drift. The union o
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  ]
},

/* ==================== G10 ==================== */
{ label:'Attribute Vectors & Hierarchical VAEs', maxTries:3,
  questions:[
    { type:'cloze', x:5, y:10, w:62, fs:11, lineSpacing:1.8,
      parts:['<b>Q25.</b> Mean latent of smiling faces = (2, 1); of non-smiling faces = (
             ').<br>Adding it to a neutral face z = (0.2, 0.3) gives (', num(1.7,0.01,45)
    { type:'radio', x:5, y:29, w:62, fs:11,
      question:'<b>Q26.</b> Which pair of factorisations defines a <i>Markovian</i> hiera
      choices:['$p=p(z_T)\\,p_\\theta(x|z_1)\\prod_{t=2}^{T}p_\\theta(z_{t-1}|z_t)$; $\\
               '$p=p(z_1)\\prod_t p_\\theta(z_t|z_{1:t-1})\\,p_\\theta(x|z_{1:T})$; $\\ q=
               '$p=p(z_T)\\,p_\\theta(x|z_{1:T})$; $\\ q=q_\\phi(z_T|x)$',
               '$p=\\prod_t p(z_t)\\,p_\\theta(x|z_T)$; $\\ q=q_\\phi(z_1|x)\\prod_t q_\\
      ans:0 },
    { type:'cloze', x:5, y:62, w:62, fs:11, lineSpacing:1.7,
      parts:['<b>Q27.</b> (a) An MHVAE with $T=1$ reduces to the vanilla VAE ELBO. ', tf(
             '<br>(b) In a variational diffusion model the encoder $q(z_t|z_{t-1})$ is lea
             '<br>(c) NVAE generates top-down and infers with a bottom-up + top-down encod
  ],
  bubbles:[
    mkBubble('<b>Q25 — (1.5, −0.5); (1.7, −0.2).</b> Attribute vector = mean(with) − mean
    mkBubble('<b>Q26 — A.</b> Each step conditions only on its neighbour: $z_T\\to\\dots\
    mkBubble('<b>Q27 — T, F, T.</b> Luo (2022): a VDM is an MHVAE with latent dim = data d
  ]
},

/* ==================== G11 ==================== */
{ label:'β-VAE, Posterior Collapse & AR vs VAE', maxTries:3,
  questions:[
    { type:'label', x:5, y:9, w:62, fs:11,
      text:'<b>Q28.</b> Match each β in $\\mathbb E_q[\\log p(x|z)]-\\beta D_{KL}$ with i
    { type:'match', x:5, y:14, w:520, fs:10, dotSize:9, dotGap:7, colGap:40,
      left:['β = 0','β = 1','β = 4','β → ∞'],
      right:['Standard VAE: the exact ELBO','Posterior collapse: z ignored, q = prior','P
      ans:[2,0,3,1] },
    { type:'radio', x:5, y:40, w:62, fs:11,
      question:'<b>Q29.</b> You pair a VAE encoder with a powerful PixelCNN decoder and ob
      choices:['The reparameterisation trick fails for PixelCNNs',
               'The AR decoder models $x$ alone, so the cheapest ELBO optimum sets $q(z|x
               'The prior variance is too small',
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              'The encoder learning rate is too low'],
      ans:1 },
    { type:'cloze', x:5, y:68, w:62, fs:11, lineSpacing:1.5,
      parts:['<b>Q30.</b> A phone app must generate faces in real time and let users edit
             {blank:true, type:'open', w:520, h:45,
              ans:'VAE. Generation is a single decoder forward pass (very fast), whereas A
              placeholder:'Write your answer here. It will be evaluated by AI…'}] }
  ],
  bubbles:[
    mkBubble('<b>Q28 — 0→AE, 1→ELBO, 4→disentangled, ∞→collapse.</b> Higgins et al.: β>1 t
    mkBubble('<b>Q29 — B.</b> Encoding costs KL nats while a strong decoder gains little.
    mkBubble('<b>Q30 — VAE.</b> One fast decoder pass + latent codes for attribute arithme
  ]
},

/* ==================== G12 ==================== */
{ label:'Discrete VAEs, Gumbel-Softmax & dVAE vs VQ', maxTries:3,
  questions:[
    { type:'cloze', x:5, y:10, w:62, fs:11, lineSpacing:1.8,
      parts:['<b>Q31.</b> Uniform prior $1/K$, so KL $=\\log K-H(q)$. One-hot $q$, $K=512$
             ' nats; uniform $q$: ', num(0,0.001,45), '<br>$K=4$, $q=(0.5,0.5,0,0)$: ', n
    { type:'cloze', x:5, y:28, w:62, fs:11, lineSpacing:1.8,
      parts:['<b>Q32.</b> Gumbel-max: $\\log f=(\\ln0.5,\\ln0.3,\\ln0.2)$, noise $g=(0.1,
             '<br>The same noise with Gumbel-softmax at $\\tau\\to0$ gives a one-hot at c
    { type:'cloze', x:5, y:48, w:62, fs:11, lineSpacing:1.9,
      parts:['<b>Q33.</b> (a) As $\\tau\\to\\infty$, each component of a $K=4$ Gumbel-soft
             '<br>(b) The DALL·E-1 dVAE discretises via ', sel(['nearest neighbour','Gumbe
             '; VQ-VAE sends gradients to the encoder via ', sel(['straight-through','sco
             '<br>(c) In VQ-VAE training the KL term is usually ', sel(['dropped','modell
  ],
  bubbles:[
    mkBubble('<b>Q31 — 6.238; 0; 0.693.</b> ln 512 = 9 ln 2; entropies 0, ln K, ln 2.',10
    mkBubble('<b>Q32 — 2; 2.</b> Scores −0.593, −0.204, −1.409 → argmax 2. At τ→0 softmax
    mkBubble('<b>Q33 — 0.25; Gumbel; ST; dropped.</b> Jang et al.: uniform as τ→∞, one-ho
  ]
},

/* ==================== G13 ==================== */
{ label:'VQ-VAE: Layer, Losses & Gradients', maxTries:3,
  questions:[
    { type:'ordering', x:5, y:10, w:62, fs:11,
      question:'<b>Q34.</b> Order the VQ-layer operations (slide 74), forward then backwa
```

```
      items:['argmin over K → indices $(n\\cdot h\\cdot w,1)$',
             'Reshape $z_e$: $(n,h,w,d)\\to(n\\cdot h\\cdot w,d)$',
             'Backward: copy $\\nabla_{z_q}L$ to $z_e$ (straight-through)',
             'Distances to all $K$ codebook vectors $(n\\cdot h\\cdot w,K)$',
             'Reshape back to $z_q$: $(n,h,w,d)$',
             'Look up codebook vectors $e_k$ by index'],
      ans:[1,3,0,5,4,2] },
    { type:'label', x:5, y:51, w:62, fs:11,
      text:'<b>Q35.</b> Which parameters does each VQ-VAE loss term update? (many-to-one a
    { type:'match', x:5, y:56, w:520, fs:10, dotSize:9, dotGap:7, colGap:40,
      left:['$\\log p(x|z_q(x))$ (reconstruction)','$\\|\\mathrm{sg}[z_e(x)]-e\\|^2$ (cod
      right:['Encoder only','Decoder and encoder','Codebook only','Nothing (constant)'],
      ans:[1,2,0,3] }
  ],
  bubbles:[
    mkBubble('<b>Q34 — reshape → distances → argmin → lookup → reshape → copy grads.</b> a
    mkBubble('<b>Q35 — dec+enc, codebook, encoder, nothing.</b> van den Oord et al.: codel
  ]
},

/* ==================== G14 ==================== */
{ label:'VQ-VAE Shapes, Priors & Dead Codes', maxTries:3,
  questions:[
    { type:'cloze', x:5, y:10, w:62, fs:11, lineSpacing:1.8,
      parts:['<b>Q36.</b> $n=8$ images, latent grid 32×32, $d=64$, $K=512$. Flattened $z_(
             ' rows; distance matrix 8192 × ', num(512,0,55), '<br>One image costs ', num
             ' bits of indices, a ', num(42.7,0.2,55), '× compression of a 128×128×3 8-bi
    { type:'radio', x:5, y:29, w:62, fs:11,
      question:'<b>Q37.</b> Why is a PixelCNN prior over the indices fitted <i>after</i> V
      choices:['The uniform prior was only a training device (constant KL); independent u
               'PixelCNN cannot be trained jointly with any network',
               'The codebook is discarded after training',
               'The uniform prior makes the decoder non-deterministic'],
      ans:0 },
    { type:'cloze', x:5, y:63, w:62, fs:11, lineSpacing:1.5,
      parts:['<b>Q38.</b> Only 40 of 512 codes are ever selected during training. Explain
             {blank:true, type:'open', w:520, h:45,
              ans:'A code is only updated (codebook loss or EMA) when it is the nearest n
              placeholder:'Write your answer here. It will be evaluated by AI…'}] }
  ],
  bubbles:[
    mkBubble('<b>Q36 — 8192; 512; 9216; ≈42.7.</b> 8·32·32 rows; 1024 indices × 9 bits; 3
```

```
    mkBubble('<b>Q37 — A.</b> The true distribution over indices is far from uniform and
    mkBubble('<b>Q38 — model answer.</b> Only the nearest code is pulled, so far-initiali
  ]
}

];
</script>

<script>
function loadEngine(){ var s=document.createElement('script'); s.src='youtube-quiz-engine
if(document.readyState==='complete'){ setTimeout(loadEngine,100); }
else { window.addEventListener('load', function(){ setTimeout(loadEngine,100); }); }
</script>
</body>
</html>
```

Groups and question count (38 questions, 14 groups, maxTries 3 each)

#	Group	Questions	Types
1	GMM & Latent Variable Basics	Q1–Q3	radio, T/F select, numeric
2	VAE as an Infinite Mixture	Q4–Q6	radio, numeric, checkbox
3	Evaluating p(x): MC & Importance Sampling	Q7–Q9	numeric, radio ×2
4	ELBO, Jensen & the KL Gap	Q10–Q12	radio, numeric, checkbox
5	EM-like Bounds & Stochastic VI	Q13–Q14	radio, ordering
6	Reparameterization Trick	Q15–Q17	numeric, radio, open (AI)
7	Amortized Inference & Amortization Gap	Q18–Q19	radio, match (many-to-one)
8	ELBO Decomposition & Closed-Form KL	Q20–Q22	numeric, slider, T/F select
9	Reading the Latent-Space Figures	Q23–Q24	match (many-to-one), radio
10	Attribute Vectors & Hierarchical VAEs	Q25–Q27	numeric, radio, T/F select
11	β-VAE, Posterior Collapse & AR vs VAE	Q28–Q30	match, radio, open (AI)
12	Discrete VAEs, Gumbel-Softmax & dVAE vs VQ	Q31–Q33	numeric ×2, numeric + select
13	VQ-VAE: Layer, Losses & Gradients	Q34–Q35	ordering, match
14	VQ-VAE Shapes, Priors & Dead Codes	Q36–Q38	numeric, radio, open (AI)

Difficulty mix:

- **Easy recall (Q1, Q2, Q10):** reading slide content directly.
- **Medium (e.g. Q4, Q14, Q23, Q34, Q35):** interpreting figures and equations.
- **Hard transfer and deduction (the rest):**
  - New numbers: KL values, gradient chains, Gumbel-max draws, tensor shapes, bit budgets.
  - New scenarios: dead codes, a mobile app, KL ≈ 0 with PixelCNN, tiny-dot ellipses.
  - Facts not on the slides: GMM parameter counting; the hidden variance bound (1, e) in slide 43's decoder; MSE+KL ⇔ $s^2 = 0.5$; zero-variance importance sampling; the amortization vs approximation gap.

How the answers were checked

Every numeric answer was recomputed by hand:

- **Gaussian KL:** uses the closed form from Kingma & Welling (AEVB, Appendix B), $-D_{KL} = \frac{1}{2}\Sigma(1 + \log\sigma^2 - \mu^2 - \sigma^2)$. arxiv
- **Reparameterisation:** uses $z = \mu + \sigma \odot \varepsilon$ (Section 3 of the same paper), so $\partial z/\partial\sigma = \varepsilon$. arxiv
- **VQ-VAE facts:** van den Oord et al. state:
  - the KL is "constant and equal to log K"; arxiv
  - the decoder optimises the reconstruction term, the encoder optimises the first and last terms, and the embeddings optimise the middle term; arxiv
  - they used $\beta = 0.25$ (results did not vary for β from 0.1 to 2.0); arxiv
  - the compression is "≈ 42.6 in bits" for 128×128×3 images mapped to 32×32 latents with K = 512 (§4.2; the exact ratio 393,216/9,216 is 42.67, so the quiz accepts 42.7 ± 0.2).
- **Gumbel-Softmax:** Jang et al. state that samples become one-hot as $\tau \to 0$ and uniform as $\tau \to \infty$, and that low τ means high-variance gradients. arxiv

Things to watch when you load it:

- **Many-to-one matching (Q19, Q23):** check that your engine version accepts repeated indices in `ans`. Q23 maps both "KL only" and "$\beta \to \infty$" to the same plot.
- **Line breaks inside cloze `parts`:** these use `<br>`, which assumes the engine renders cloze parts as HTML, as the kNN template does.
- **Q38 ("dead codes"):** the original VQ-VAE paper only mentions moving-average codebook updates, and it did not use them. The dead-code problem is described by Yu et al. (ViT-VQGAN, Google Research), who write that "Vanilla VQVAEs usually suffer from low codebook usage due to the poor initialization of the codebook… a significant portion of codes are rarely used, or dead." They fix it with factorised low-dimensional codes and $\ell 2$-normalised codes. Restarting dead codes and EMA updates are later common practice. Grade Q38 as a reasoning question rather than against one canonical answer.

External sources used

- Stanford CS236 (Ermon), Lecture 6 slides: amortized inference, SVI, the "Problem!" gradient. github
- Kingma & Welling, Auto-Encoding Variational Bayes: Appendix B KL, Section 2.4 reparameterisation. arxiv
- Kingma & Welling, An Introduction to Variational Autoencoders: ELBO tightness. arxiv-vanity

- van den Oord, Vinyals & Kavukcuoglu, Neural Discrete Representation Learning (VQ-VAE). arxiv
- Razavi, van den Oord & Vinyals, VQ-VAE-2: two-stage training, hierarchical PixelCNN priors. arxiv
- Jang, Gu & Poole, Categorical Reparameterization with Gumbel-Softmax. arxiv
- Higgins et al. β-VAE, via Lilian Weng, "From Autoencoder to Beta-VAE"; Burgess et al. on capacity. github
- Calvin Luo, Understanding Diffusion Models: A Unified Perspective (MHVAE, VDM). arxiv
- Cremer, Li & Duvenaud, Inference Suboptimality in VAEs (amortization vs approximation gap); Kim et al., semi-amortized VAEs. arxiv arxiv
- Jeffares & Liu, An Introduction to Discrete VAEs (arXiv 2505.10344); UW CSE599i (Thickstun) VAE and discrete-VAE notes. arxiv washington
- Murphy, PML: Advanced Topics ch. 21 (VAE, posterior collapse, VQ-VAE, learning the prior). ublish dokumen
- Posterior-collapse literature: Bowman et al. KL annealing, free bits (He et al. 2019; Li et al. 2019). arxiv arxiv
- Ramesh et al. (OpenAI), Zero-Shot Text-to-Image Generation (DALL·E 1). The dVAE compresses "each 256×256 RGB image into a 32×32 grid of image tokens, each element of which can assume 8192 possible values", trained with the Gumbel-softmax relaxation (τ annealed to 1/16, KL weight $\beta = 6.6$).
- Yu et al. (Google Research), ViT-VQGAN: dead codes from poor codebook initialisation, fixed by factorised and $\ell 2$-normalised codes.